

EDUCATION	
Birla Insitute of Technology Mesra	Ranchi, India
Integrated Master's of Science - Mathematics & Computing; GPA: 8.88	2019 - 2024

PUBLICATIONS	
Cook-Gen: Robust Generative Modeling of Cooking Actions from Recipes at IEEE SMC 2023. Spatial Field Fusion Network (SFFNet) for Panoramic Dental X-ray Segmentation at IEEE APSCON 2023. K-PERM: Personalized Response Generation Using Dynamic Knowledge Retrieval and Persona-Adaptive Queries at AAAI Spring Symposium 2024.	

EXPERIENCE	
Thomson Reuters Lab	Bangalore, India
Applied Research Scientist	Aug 2024 - Current
- o Legal AI Reasoning and Model Enhancement: * Pioneered legal synthetic data generation pipeline for Process Reward Models (PRMs), improving LegalBench benchmark performance; evaluated Test-Time Scaling paradigms for optimizing reasoning inference. * Architected IRAC Knowledge Graph framework using Westlaw corpus, generating high-quality preference datasets for fine-tuning LLMs with improved legal reasoning alignment.	
- o AI-Powered Legal Document Update System: * Developed a comprehensive LLM workflow for automated Word document updates using XML parsing and contextual mapping, integrated with human-in-the-loop validation featuring reasoning chains, achieving 70%+ automation success and 60% manual time reduction while maintaining legal accuracy. * DocEvolver: Created an MVP for “Cursor for Word” as an extension for updating and understanding MS Word Files for Lawyer-Editors	
- o Search-and-Replace Agentic System: * Architected multi-agent AI system for automated Word editing with error-resolving agents using tool calling and multi-turn reasoning, achieving 98% accuracy and 65% reduction in manual revision time; processed hundreds of documents monthly with sub-8 second latency per section * Designed GRPO-based fine-tuning pipeline for Qwen2.5-7B with custom reward functions to enhance document editing capabilities for Search and Replace Tool.	
Microsoft Research	Bangalore, India
Research Intern (LLMs & AI4Code)	Jan 2024 - July 2024
- o Programming with Representations (PwR): * Led development for Microsoft’s PwR Studio platform that enables non-programmers to create applications through natural language; built NL2DSL translation system using GPT-4, achieving 85% hallucination reduction and improving valid code generation from 65% to 95% * Architected Jugalbandi platform that converts DSL into production-ready chatbot applications, reducing development time by 80%; enabled 15+ non-technical organizations to deploy AI-powered conversational bots	
o Multimodal Chatbot Infrastructure: Developed end-to-end management system supporting multilingual voice/text interactions across WhatsApp, Telegram, and web; integrated Bhashini Speech models with Azure failover for 70% faster deployment	
o Open Source Impact: PwR-Studio, PwR-NL2DSL, Jugalbandi Studio Engine	
AI Institute, University of South Carolina	Remote
Research Intern (Neuro-Symbolic AI & NLP)	Dec. 2022 - April 2024
o Master thesis on Knowledge Enabled Multimodal Ingredient Substitution, built knowledge graph incorporating 27K ingredients and 80K substitution pairs. Enabled precise ingredient recommendations using multimodal and constraint-based searches. We built an LLM-based query module for the ingredient substitution knowledge graph. We also submitted our work to AAAI-25.	
Saarland University	Germany
Visiting Researcher	May 2023 - Aug 2023
o Conducted research in Satellite Image Superresolution, enhancing disaster analysis through high-temporal frequency & low-resolution images using GAN & Diffusion models under Dr. Ingmar Weber and Dr. Ferda Ofli.	

SKILLS	
General	Python, C/C++, SQL, Docker, Kafka, Git/GitHub, AWS, Selenium, BeautifulSoup
Web Technologies	HTML/CSS, Flask, JavaScript, Beego/Gin(Golang)
Deep Learning & ML	PyTorch, Tensorflow, Scikit-Learn, OpenCV, OpenVino, TensorRT, SpaCy
Generative AI	Large Language Models, RAG, LlamaIndex, LangChain, Agentic Systems